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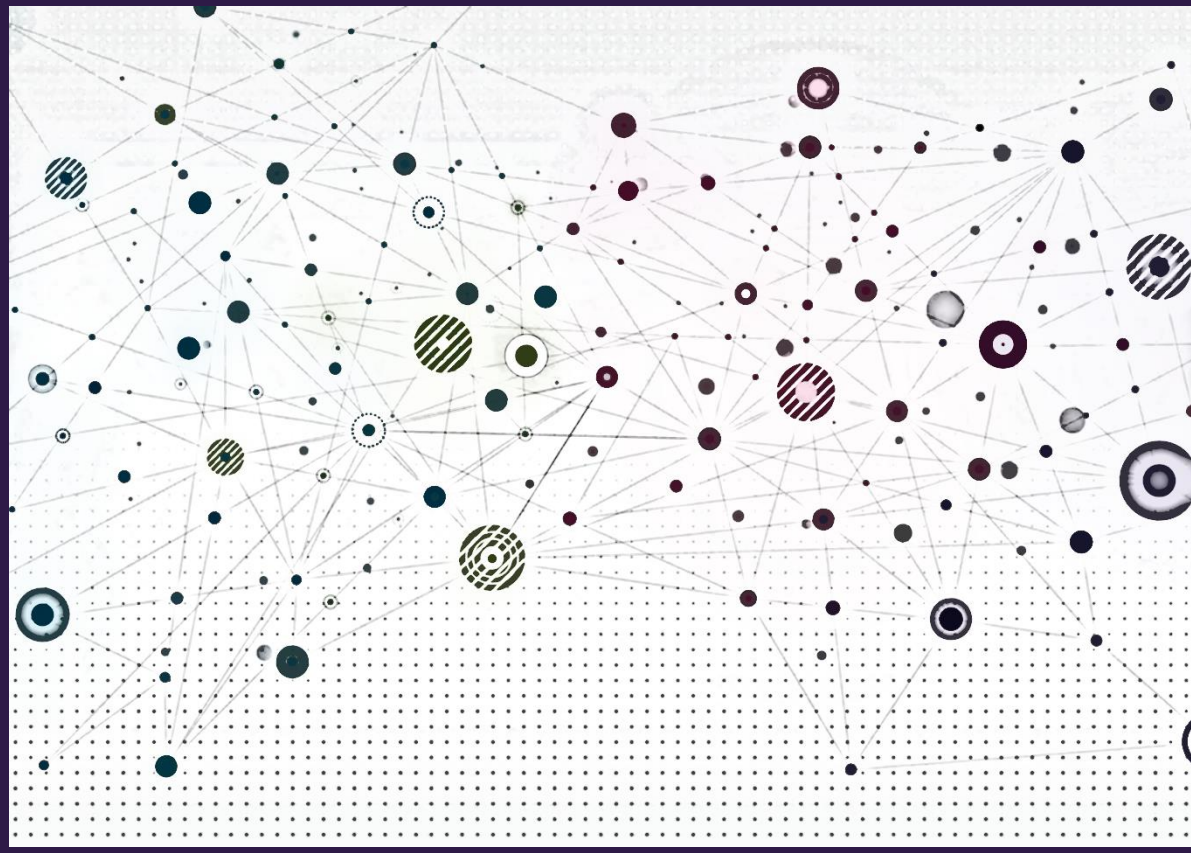
Shallow Node Embeddings



Overview

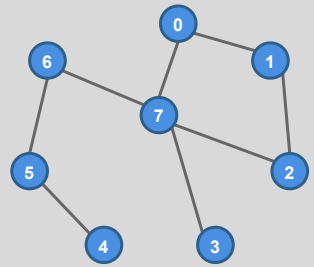
- Motivation
- Shallow node embedding framework
- Node embedding methods
  - Factorization approaches
  - Random walk approaches
- Graph embedding
- Limitations of matrix factorization and random walk embeddings

Motivation

Previous lab – Machine learning with engineered features

- In a standard machine learning approach, we extract mathematically formulated metrics from an input graph to train and make inferences with a statistical learning model
- As with many other areas (e.g., computer vision, natural language processing), it is frequently beneficial to automate feature learning and bypass feature engineering

Input graph



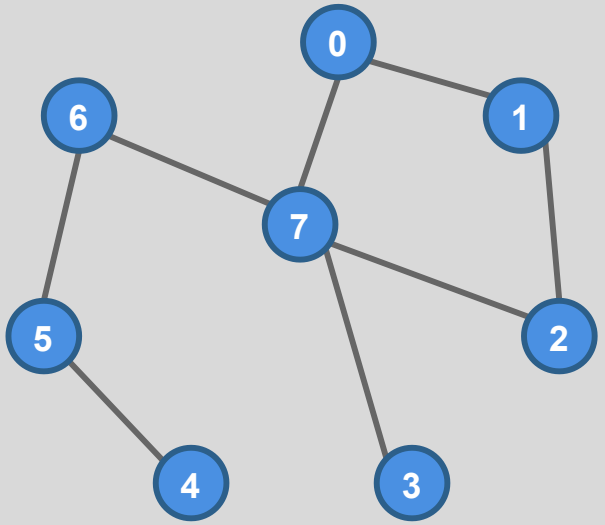
Engineered Features (centrality, cliques, k-cores, diameter, etc)

Machine learning algorithm

Inference

Latent representation learning for graphs

We would like to learn node, edge, and graph representations (embeddings) that are useful for many tasks



$f: u \rightarrow \mathbb{R}^d$

Vector in $\mathbb{R}^d$

- f is a function that maps an input, u, (node, edge, or graph) to numerical vector in $\mathbb{R}^d$
- In this course, f will typically be a neural network

Why learn embeddings?

- Eliminates reliance on handcrafted features
- May capture network information that is not in engineered features
- Potentially useful for many downstream tasks
- Pretraining with self-supervised learning can reduce labeled data requirements for downstream tasks

Embedding in $\mathbb{R}^d$

Pre-trained node or graph via self-supervised learning

Downstream Tasks

- Node classification
- Graph classification
- Clustering
- Graph generation
- … and so on

Example node embeddings

Image credit: Perozzi 2014

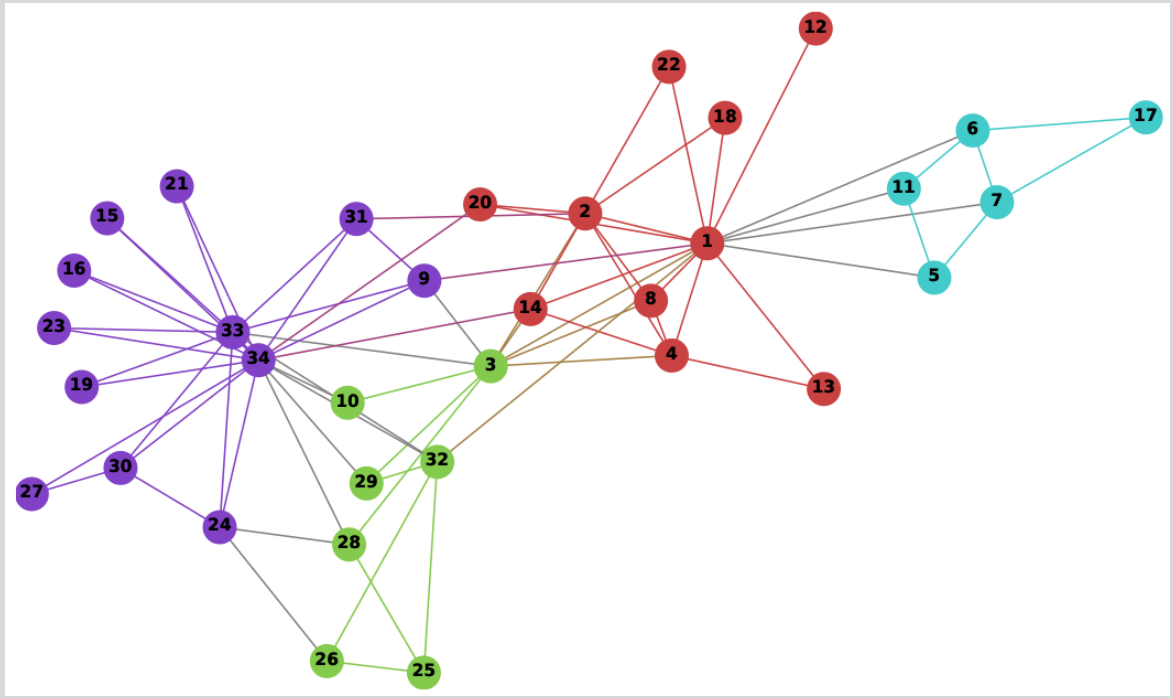
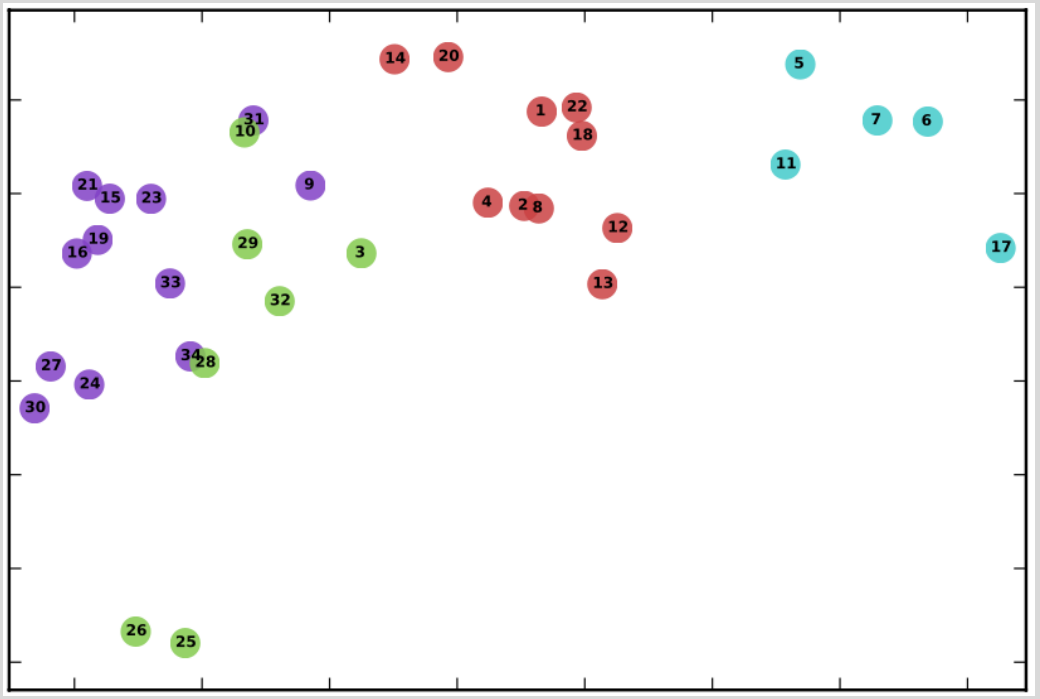


Image credit: Perozzi 2014



Embeddings in $\mathbb{R}^2$ for Zachery's Karate network. Notice how community structure of the graph is maintained by the embeddings.

Shallow Node Embedding Framework

Encoder-decoder conceptual description

- We will adopt an encoder-decoder paradigm where:
  - Encoder maps each node (or edge) to an embedding, i.e., a vector in $\mathbb{R}^d$
  - Decoder takes one or more embeddings and infers information about a node, edge, or graph.
  - Typically, we will only care about the encoder for downstream tasks, but both are necessary to form the embeddings

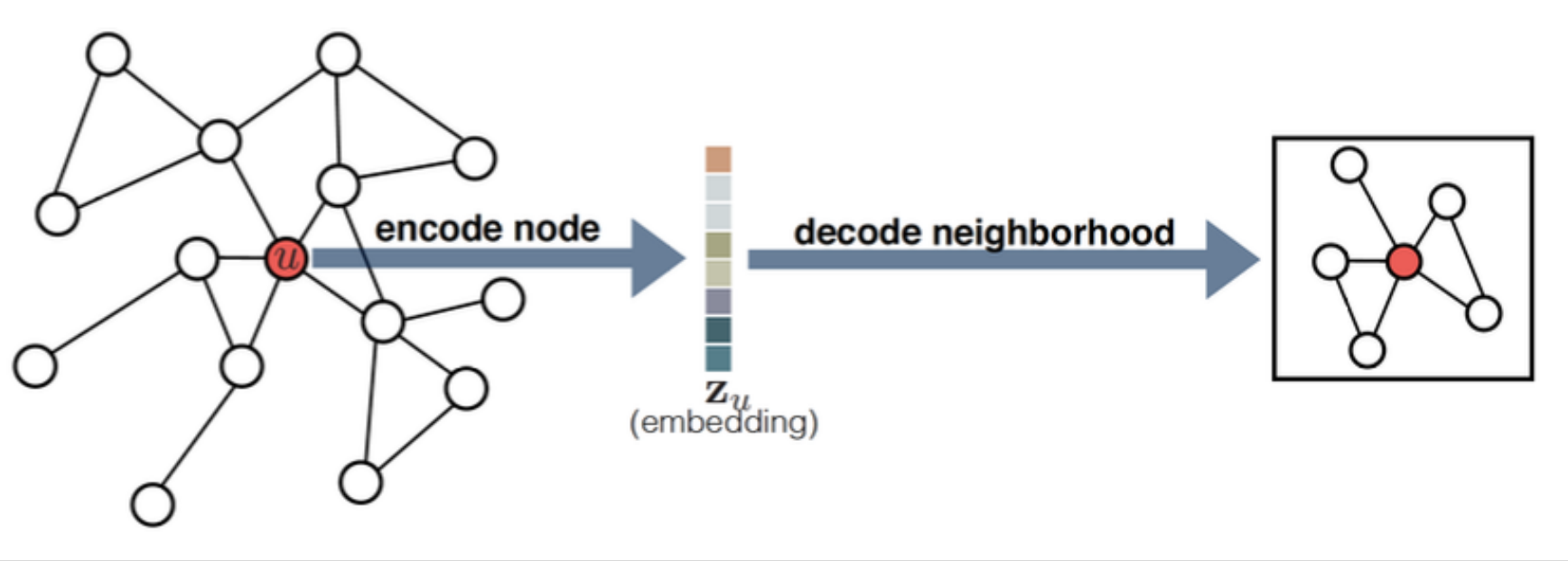


Image credit: Hamilton 2020

Encoder-decoder components

- The Encoder, ENC, is a function (e.g., a neural network) that maps nodes to an $\mathbb{R}^d$ embedding
- A node (or edge) similarity measure, $\mathbf{S}[u, v]$, that we'll use to assess performance of the encoder-decoder model
- The Decoder, DEC, infers the similarity measure from embedding inputs
- Optimization approach to learn model parameters so that:

$$DEC(\boldsymbol{e}_u, \boldsymbol{e}_v) \approx \mathbf{S}[u, v] \text{ for all nodes (or edges) in the network}$$

$$DEC\big(ENC(u), ENC(v)\big) \approx \mathbf{S}[u, v] \text{ for all nodes (or edges) in the network}$$

Encoder concept in detail

- The *encoder*, $ENC(v) = \mathbf{z}_v$, is any function (e.g., a neural network) that maps a node, v, to a vector, $\mathbf{z}_v \in \mathbb{R}^d$
- d is often called the embedding dimension or size
- Today, we will focus on shallow methods that only encode network structure
- Later, we'll see graph neural networks that also use node features

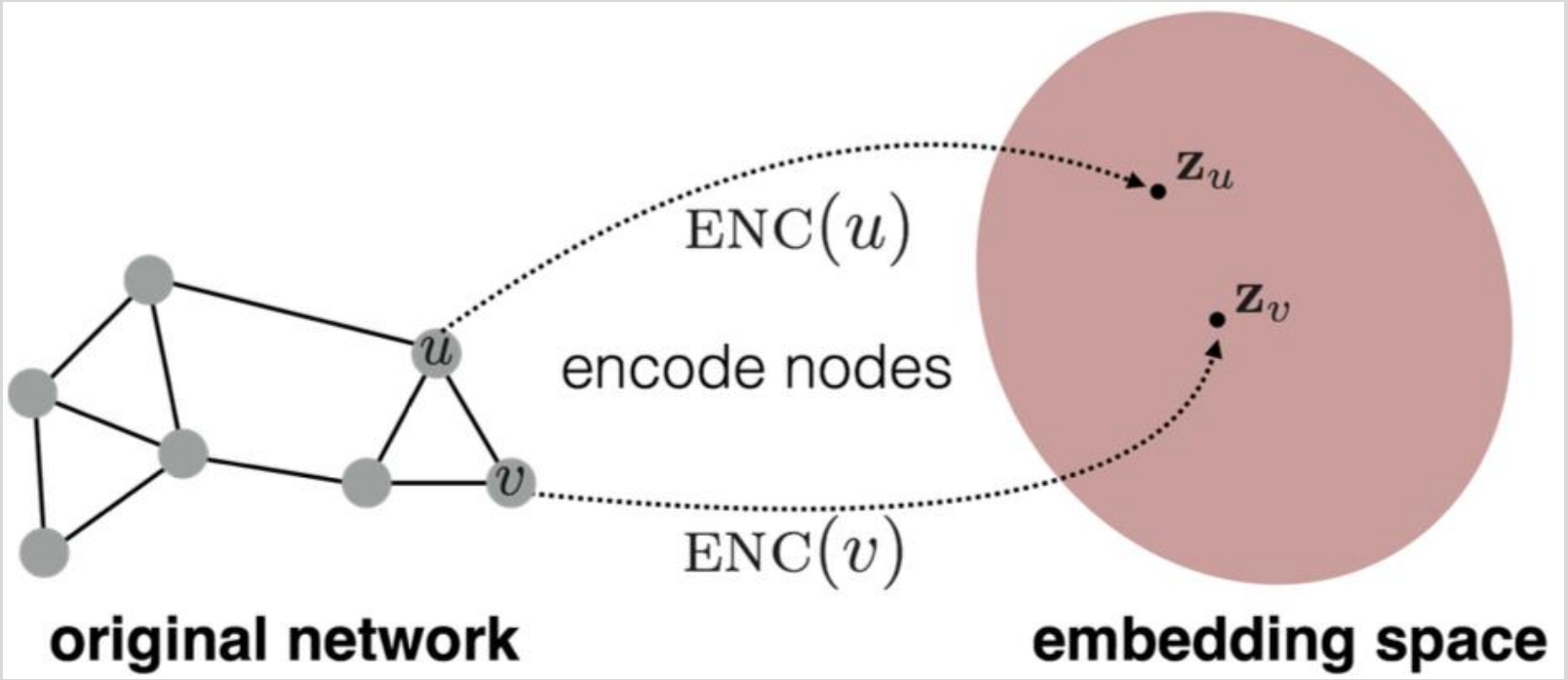
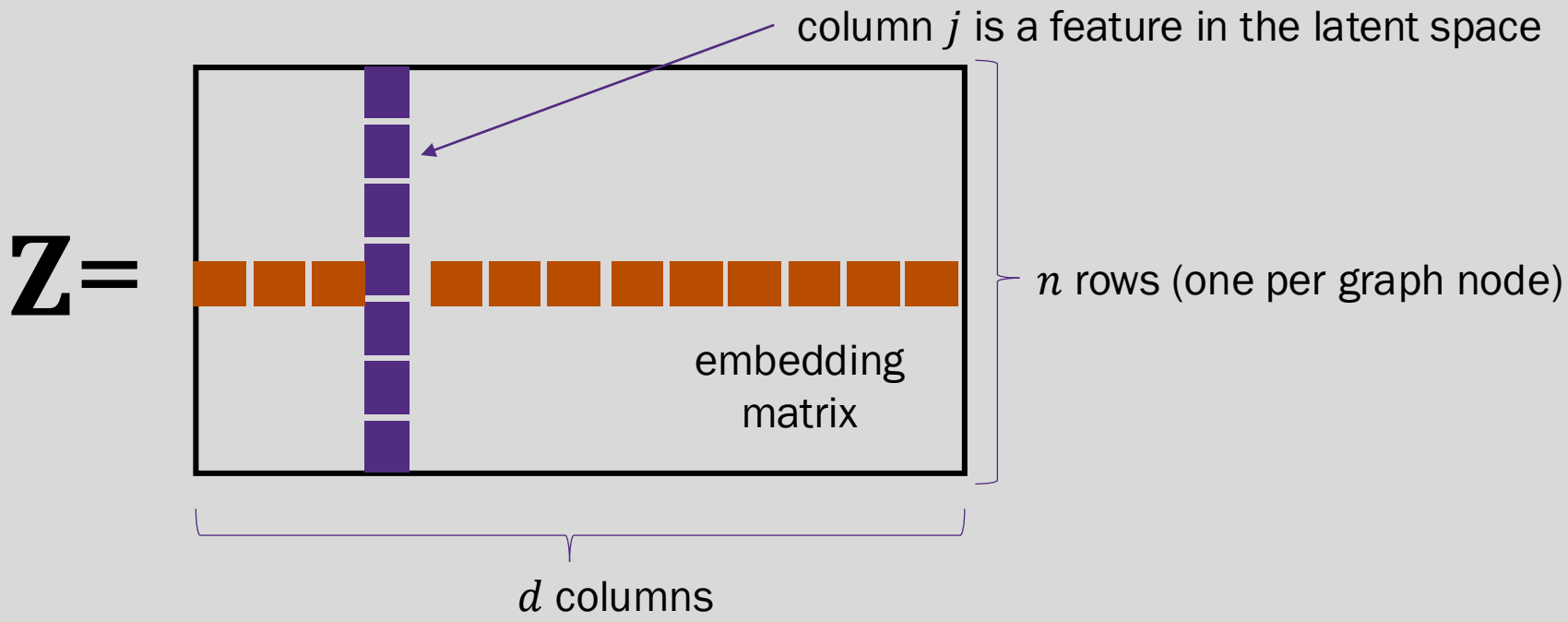


Image credit: Hamilton 2020

Encoder as a lookup function

- In the simplest case:
  - The graph is static (no nodes added or removed)
  - The embedding is constant for every node and stored in an embedding matrix, $\mathbf{Z} \in \mathbb{R}^{n \times d}$ where d is the embedding dimension and n is the number of nodes
  - The encoder is just a lookup, i.e., given a node ID, i, it returns the i^{th} row from $\mathbf{Z}$



Embedding learning through loss optimization

- We consider a **pairwise decoder**, $DEC(\mathbf{z}_u, \mathbf{z}_v)$, that is a function of any two node embedding inputs
- Let $\mathbf{S}[u, v] \in \mathbb{R}^+$ be a similarity measure of nodes u and v
- Our goal is to construct an encoder-decoder model such that

$$DEC(\mathbf{z}_u, \mathbf{z}_v) \approx \mathbf{S}[u, v]$$

- Standard approach is to minimize the empirical loss, $\mathcal{L}$, over a set of training node pairs $\mathcal{D}$:

$$\mathcal{L} = \sum_{(u,v)\in\mathcal{D}} l\big[DEC(\mathbf{z}_u, \mathbf{z}_v), \mathbf{S}[u, v]\big]$$

  where $l \in \mathbb{R} \times \mathbb{R} \longrightarrow \mathbb{R}$ is a loss function
- We will discuss specific similarity measures and loss functions in the following slides

Shallow node embedding qualities

- Unsupervised or self-supervised learning task (depends on specific method)
- Node features are not used
- Node labels are not used
- Goal is to learn latent features that the decoder can use to reconstruct a similarity measure derived from network structure
- The node embeddings are intended to be useful for any task where network structure is informative

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Node Embedding with Matrix Factorization

Matrix factorization embeddings

- Simplest node similarity: Nodes u and v are similar if they are connected by an edge
- Let the decoder be $DEC(\mathbf{z}_u, \mathbf{z}_v) = \mathbf{z}_v^T \mathbf{z}_u$ (inner product decoder)
- This means we want : $\mathbf{z}_v^T \mathbf{z}_u = a_{u,v}$ which is the (u, v) entry of the graph adjacency matrix $\mathbf{A}$
- Equivalently, letting $\mathbf{Z} \in \mathbb{R}^{n \times d}$ be the embedding matrix, we want $\mathbf{Z}\mathbf{Z}^T = \mathbf{A}$

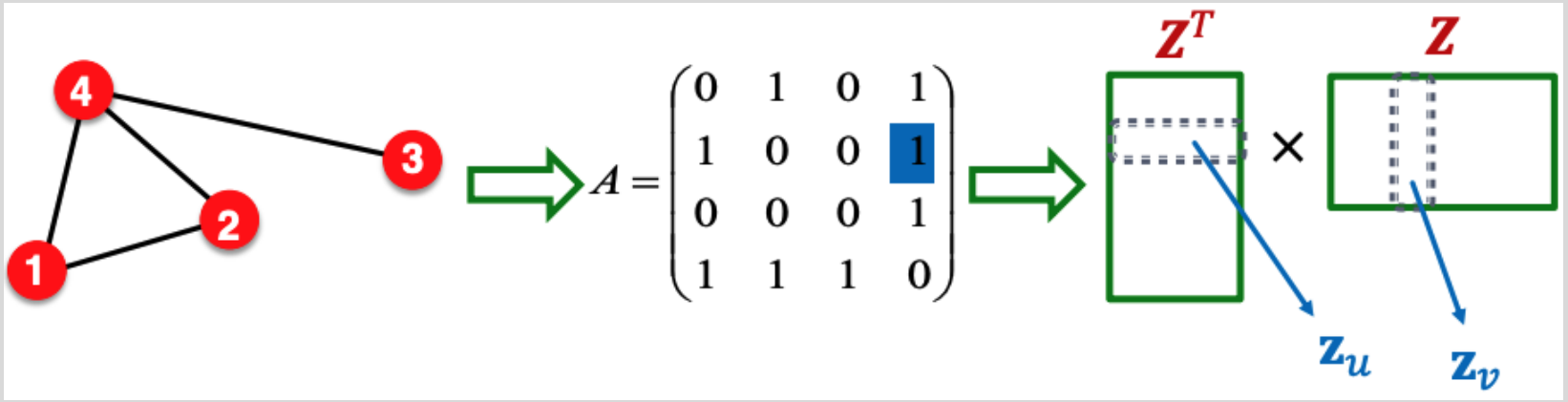

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Matrix factorization embeddings

- The embedding dimension d (number of columns in $\mathbf{Z}$) is much smaller than number of nodes n
- Exact factorization $\mathbf{Z}\mathbf{Z}^T = \mathbf{A}$ is generally not possible
- However, we can learn $\mathbf{Z}$ approximately
- Objective function: $\min_{\mathbf{Z}} \|\mathbf{A} - \mathbf{Z}\mathbf{Z}^T\|_2$ - here the similarity function is the adjacency matrix, $\boldsymbol{S} = \mathbf{A}$
  - We optimize $\mathbf{Z}$ such that it minimizes the L2 norm (Frobenius norm)

$$\|\mathbf{A} - \mathbf{Z}\mathbf{Z}^T\|_2 = \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n}\left(a_{ij} - \left(\mathbf{z}_i^T\mathbf{z}_j\right)\right)^2}$$

- Inner product decoder with node similarity defined by edge connectivity is equivalent to matrix factorization of $\mathbf{A}$

Matrix factorization embeddings

- Other methods adopt the same approach for matrix factorization embeddings
  - Same inner product decoder $DEC(\mathbf{z}_u, \mathbf{z}_v) = \mathbf{z}_v^T \mathbf{z}_u$
  - Same objective function $\min_{\mathbf{Z}} \|\mathbf{S} - \mathbf{Z}\mathbf{Z}^T\|_2$
- The difference is how the similarity, $\mathbf{S}$, is defined (there are many alternatives)

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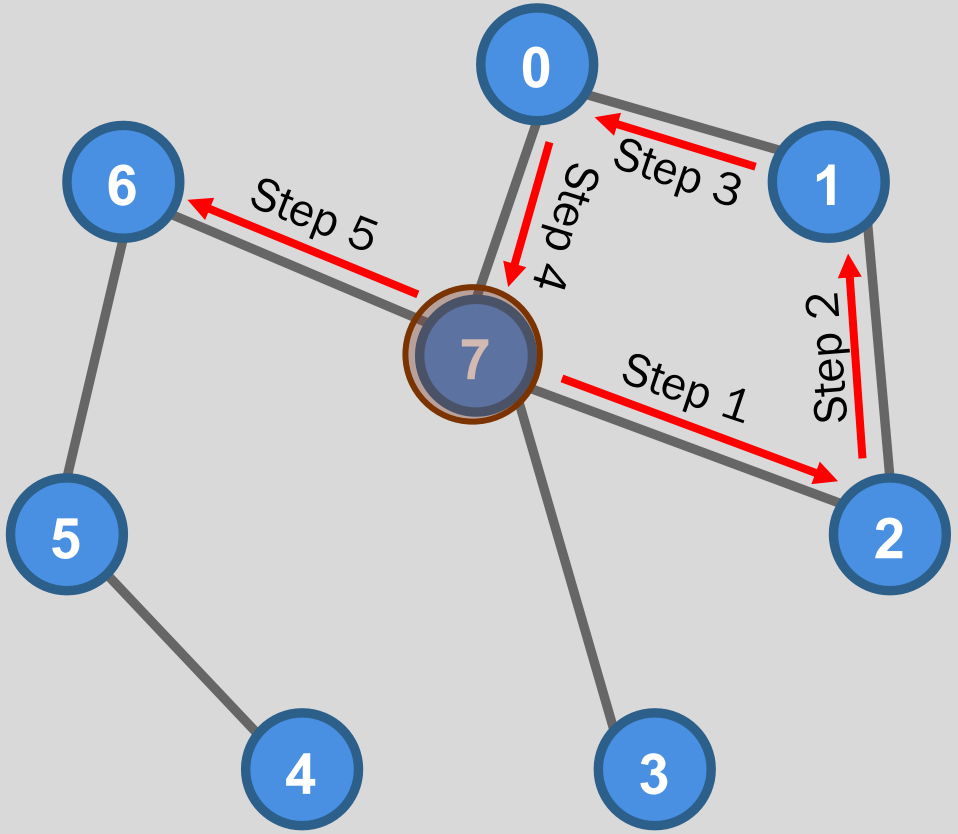
Node Embedding with Random Walks

Random walk



Starting at node 7, and taking 5 random steps yields the walk:

$$7 \rightarrow 2 \rightarrow 1 \rightarrow 0 \rightarrow 7 \rightarrow 6$$

1. Start at a randomly selected node, u. Set $N = n(u)$ to the neighbors of u
2. Randomly pick a node v from N.
3. Set $u = v$
4. Set $N = n(u)$
5. Repeat 2-4 for $T - 1$ times where T is the desired path length (edge count)

- Loops are allowed, including going back to the previous node from the current node

Random walk embedding concept

- Intuition: Two nodes that co-occur on short random walks are likely to be structurally similar (share neighbors) and should have similar embeddings
- We can express this intuition mathematically by setting the similarity function, $\mathbf{S}[u, v]$, to

$$\mathbf{S}[u, v] \equiv p_{G,T}(v|u)$$

where $p_{G,T}(v|u)$ is the probability of visiting node v on a length-T random walk starting at node u on graph G

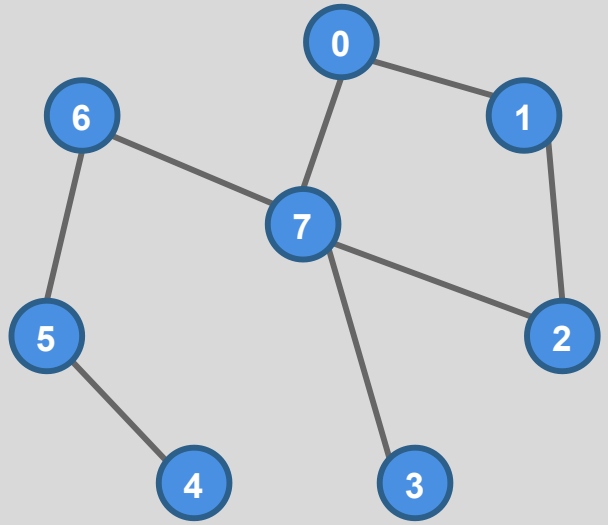
- Our goal is to develop an encoder-decoder model such that

$$DEC(\mathbf{z}_u, \mathbf{z}_v) \approx p_{G,T}(v|u)$$

for any pair of nodes u and v with embeddings $\mathbf{z}_u$ and $\mathbf{z}_v$

Random walk similarity measure

- In the previous slide, we set $\mathbf{S}[u, v] \equiv p_{G,T}(v|u)$
- But we don't know $p_{G,T}(v|u)$. What can we do?
- Generate many length-T random walks using a walk strategy R
- Estimate probabilities empirically from the random walks



Length-3 Random Walks
7→2→1
7→0→1
7→3→7
7→6→5
7→0→1

Probability Estimates

$p_{G,3}(0|7) = {}^{2}/_{5}$

$p_{G,3}(1|7) = {}^{3}/_{5}$

$p_{G,3}(2|7) = {}^{1}/_{5}$

$p_{G,3}(3|7) = {}^{1}/_{5}$

$p_{G,3}(4|7) = {}^{0}/_{5}$

$p_{G,3}(5|7) = {}^{1}/_{5}$

$p_{G,3}(6|7) = {}^{1}/_{5}$

Random walk decoder

- A natural choice for a decoder to approximate $p_{G,T}(v|u)$ is the softmax

$$DEC(\mathbf{z}_u, \mathbf{z}_v) = \frac{exp(\mathbf{z}_u^T \mathbf{z}_v)}{\sum_{n \in V} exp(\mathbf{z}_u^T \mathbf{z}_n)}$$

  where V is the set of all nodes in the graph

- This will result in all values of $DEC(\mathbf{z}_u, \mathbf{z}_v)$ being in $[0,1]$ and $\sum_{v \in V} DEC(\mathbf{z}_u, \mathbf{z}_v) = 1$ which can be interpreted as a probability distribution
- We want to optimize the encoder-decoder, that is we want to learn embeddings such that

$$DEC(\mathbf{z}_u, \mathbf{z}_v) = \frac{exp(\mathbf{z}_u^T \mathbf{z}_v)}{\sum_{n \in V} exp(\mathbf{z}_u^T \mathbf{z}_n)} \approx p_{G,T}(v|u)$$

Naïve random walk embedding optimization

- We could attempt to minimize the square difference

$$\left(\frac{exp(\mathbf{z}_u^T \mathbf{z}_v)}{\sum_{n \in V} exp(\mathbf{z}_u^T \mathbf{z}_n)} - \hat{p}_{G,T}(v|u) \right)^2$$

where $\hat{p}_{G,T}(v|u)$ is an empirical estimate of $p_{G,T}(v|u)$ based on random walk samples.

  - Intractable for large graphs as requires counting co-occurrences for every pair of nodes which is $O(n^2)$
  - In most networks, most pairs will never co-occur on short walks leading to 0 empirical probability which could yield large negative embeddings
- What else can we try?

Random walk embedding optimization by maximum likelihood

- Let's consider a given node u and the set of all nodes $N_R(u)$ visited on some random walk starting at u using a random walk policy R
- Applying maximum likelihood theory, we seek to minimize the objective function

$$\mathcal{L} = \sum_{u \in V} \sum_{v \in N_R(u)} -log \left[\frac{exp(\mathbf{z}_u^T \mathbf{z}_v)}{\sum_{n \in V} exp(\mathbf{z}_u^T \mathbf{z}_n)} \right]$$

Sum over all nodes u

Sum over nodes v seen on random walks starting from u

Estimated probability of u and v co-occurring on a random walk

- Optimization means finding embeddings $\mathbf{z}$ that minimize $\mathcal{L}$

Random walk embedding optimization by maximum likelihood

- Why does this work?

$$\mathcal{L} = \sum_{u \in V} \sum_{v \in N_R(u)} -log\left[\frac{exp(\mathbf{z}_u^T \mathbf{z}_v)}{\sum_{n \in V} exp(\mathbf{z}_u^T \mathbf{z}_n)}\right]$$

$$= \sum_{u \in V} \sum_{v \in N_R(u)} -log[exp(\mathbf{z}_u^T \mathbf{z}_v)] + log\left[\sum_{n \in V} exp(\mathbf{z}_u^T \mathbf{z}_n)\right]$$

- Minimizing this function requires
  - Pushing $\uparrow \mathbf{z}_u^T \mathbf{z}_v$ - requires $\mathbf{z}_u$ and $\mathbf{z}_v$ to be similar (ideally, correlated)
  - Pushing $\downarrow \sum_{n \in V} exp(\mathbf{z}_u^T \mathbf{z}_n)$ - requires $\mathbf{z}_u$ to be dissimilar (ideally, anticorrelated) from all $\mathbf{z}_n$ especially $\mathbf{z}_n \notin N_R(u)$
- Eliminates the need to count co-occurrence for every pair of nodes and addresses 0 empirical probability issues, but …

Random walk embedding optimization by maximum likelihood

- Too computationally expensive

$$\mathcal{L} = \sum_{u \in V} \sum_{v \in N_R(u)} -log \left[\frac{exp(\mathbf{z}_u^T \mathbf{z}_v)}{\sum_{n \in V} exp(\mathbf{z}_u^T \mathbf{z}_n)} \right]$$

Nested summation over all nodes in the graph gives $O(n^2)$ complexity. This is prohibitive on large graphs.

Negative sampling

Revise our loss function using the approximation*

$$-log\left[\frac{exp(\boldsymbol{z}_u^T\boldsymbol{z}_v)}{\sum_{n\in V} exp(\boldsymbol{z}_u^T\boldsymbol{z}_n)}\right]$$

$$\approx -\log[\sigma(\boldsymbol{z}_u^T\boldsymbol{z}_v)] - \sum_{i=1}^{k} \log\left[\sigma\left(-\boldsymbol{z}_u^T\boldsymbol{z}_{n_i}\right)\right], n_i \sim P_V$$

random distribution over all nodes

sigmoid function

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

- Normalize against k randomly selected *negative samples*
- A form of *Noise Contrastive Estimation (NCE)*
- Eliminates the need to compute the denominator over all nodes
- Complexity is independent of number of nodes in network

*See https://arxiv.org/pdf/1402.3722

Negative sampling

Revise our loss function using the approximation*

$$-log\left[\frac{exp(\boldsymbol{z}_u^T\boldsymbol{z}_v)}{\sum_{n\in V} exp(\boldsymbol{z}_u^T\boldsymbol{z}_n)}\right]$$

$$\approx -\log[\sigma(\boldsymbol{z}_u^T\boldsymbol{z}_v)] - \sum_{i=1}^{k} \log\left[\sigma\left(-\boldsymbol{z}_u^T\boldsymbol{z}_{n_i}\right)\right], n_i \sim P_V$$

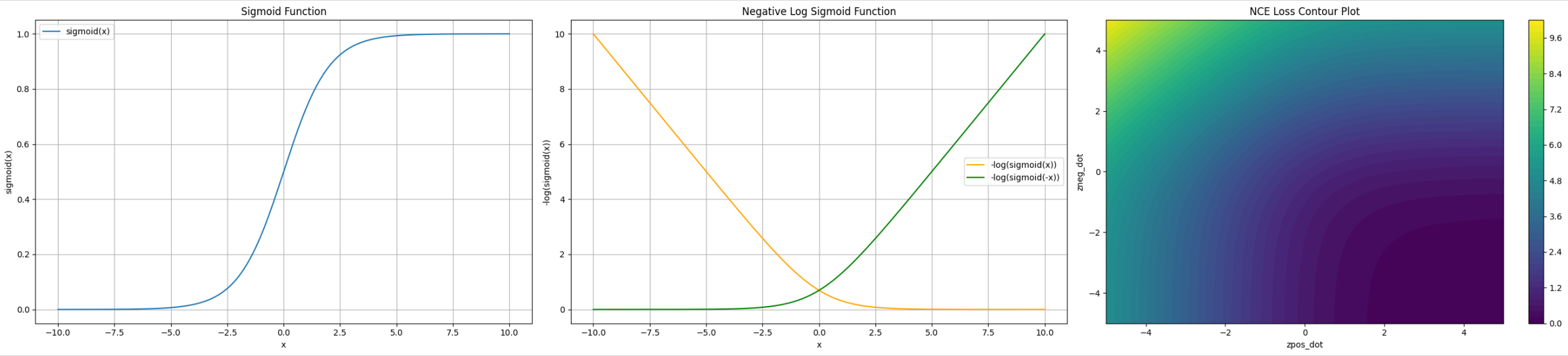
- Sample k negative nodes n_i with probability proportional to node degree
- Higher k gives more robust estimates but also higher bias on negative events (meaning, the negative sample node might actually occur on random walks we just didn't take enough samples)
- In practice, $k \in [5, 20]$

*See https://arxiv.org/pdf/1402.3722

NCE (loss function version)

$$-\left[\log[\sigma(\mathbf{z}_u^T\mathbf{z}_v)] + \sum_{i=1}^{k} \log[\sigma(-\mathbf{z}_u^T\mathbf{z}_{n_i})], n_i \sim P_V\right]$$

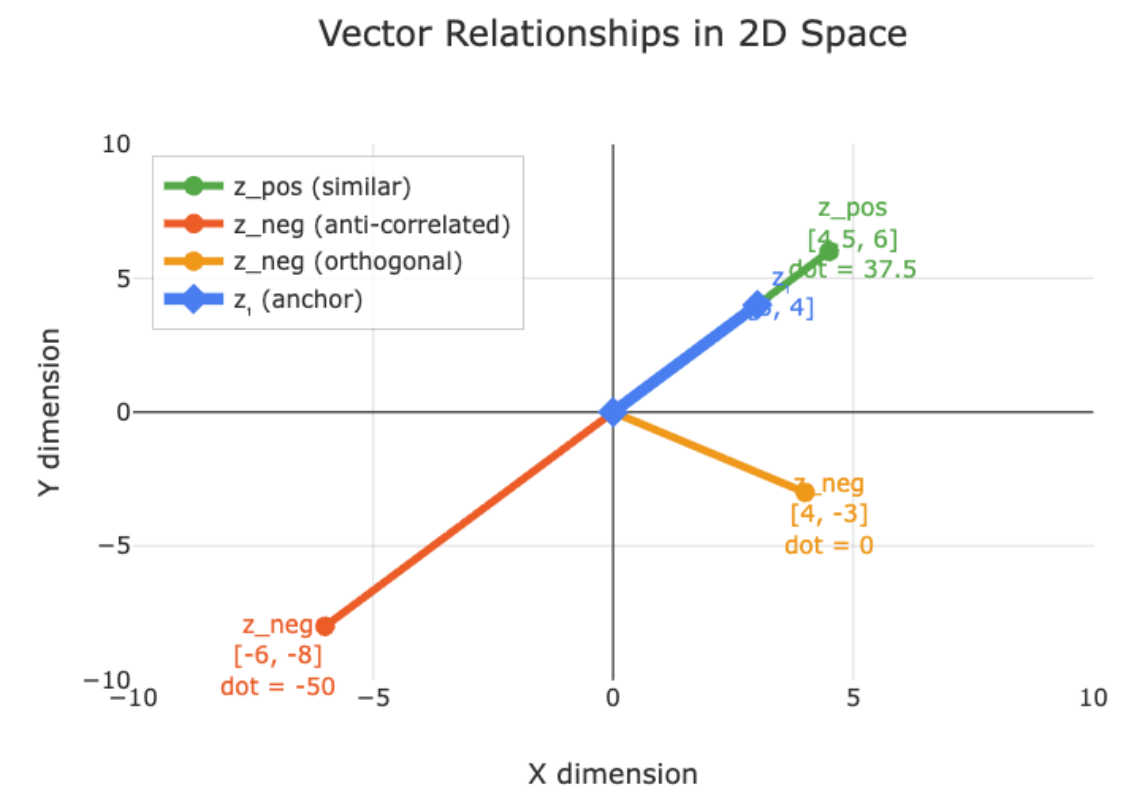
Contour plot of the NCE loss function. The x-axis is the value of $\mathbf{z}_u^T\mathbf{z}_v$. The y-axis is the value of $\mathbf{z}_u^T\mathbf{z}_{n_i}$.



The orange curve represents the value of $-\log[\sigma(\mathbf{x})]$ where $\mathbf{x} = \mathbf{z}_u^T\mathbf{z}_v$. The best we can do is make this dot product large. The green curve represents $-\log[\sigma(-\mathbf{x})]$ where $\mathbf{x} = \mathbf{z}_u^T\mathbf{z}_{n_i}$. The best we can do is make this value negative with large magnitude.

Understanding NCE loss

$$-\left[\log[\sigma(\mathbf{z}_u^T\mathbf{z}_v)] + \sum_{i=1}^{k} \log[\sigma(-\mathbf{z}_u^T\mathbf{z}_{n_i})], n_i \sim P_V\right]$$



Key Insight: Both similar vectors (positive pairs) and anti-correlated vectors (negative pairs) achieve low loss values. The model learns both *attraction* and *repulsion* relationships!

A. Similar Vectors (Positive Pair)

`z1 = [3, 4], z_pos = [4.5, 6]`

`dot_product = 3×4.5 + 4×6 =` **37.5**

`sigmoid(37.5) =` **≈ 1.000**

`log(sigmoid(37.5)) =` **≈ 0.000**

✓ *Low loss for positive pair - vectors learn to align*

B. Anti-correlated Vectors (Negative Pair)

`z1 = [3, 4], z_neg = [-6, -8]`

`dot_product = 3×(-6) + 4×(-8) =` **-50**

`sigmoid(-(-50)) = sigmoid(50) =` **≈ 1.000**

`log(sigmoid(50)) =` **≈ 0.000**

✓ *Low loss for negative pair - strong opposition is good!*

C. Orthogonal Vectors (Negative Pair)

`z1 = [3, 4], z_neg = [4, -3]`

`dot_product = 3×4 + 4×(-3) =` **0**

`sigmoid(-0) = sigmoid(0) =` **0.500**

`log(sigmoid(0)) =` **≈ -0.693**

↑ *Higher loss - neutral relationship, room for improvement*

How do we optimize our objective function?

- Assume our loss function is $\mathcal{L}$ and we want to optimize the embedding $\mathbf{z}_u$
- Stochastic Gradient Descent – process one sample at a time
  1. Randomly initialize $\mathbf{z}_m$ for all nodes m
  2. For each random walk sample $s = \{u, v_1, \cdots, v_w, n_1, \cdots, n_k\}$
     a. Compute the partial derivatives $\frac{\delta \mathcal{L}}{\delta \mathbf{z}_m}$ for all nodes $m \in s$
     b. For all nodes m, step in the negative gradient direction $z_m \leftarrow z_m - \alpha \frac{\delta \mathcal{L}}{\delta \mathbf{z}_m}$ where α is the learning rate
  3. Repeat 2 until convergence
- Completion of step 2 represents one epoch

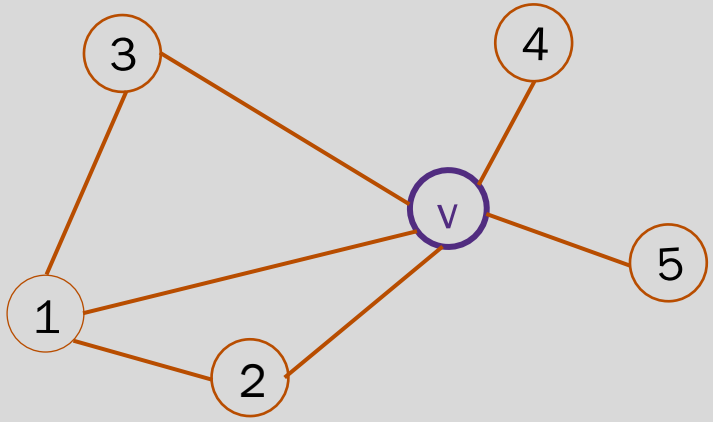
Random walk summary

1. Generate short fixed-length random walks starting from each node, u, on the graph using a random walk policy R
2. For each node u, determine the neighborhood $N_R(u) = \{v_1, \cdots, v_w\}$ from the walks and combine with k randomly selected negative nodes $\{n_1, \cdots, n_k\}$. This represents a single sample $s = \{u, v_1, \cdots, v_w, n_1, \cdots, n_k\}$
3. Optimize the embeddings $\mathbf{Z}$ using stochastic gradient descent with the NCE loss function (using negative sampling)

$$\mathcal{L}_s = \sum_{v \in N_R(u)} -log[\sigma(\mathbf{z}_u^T \mathbf{z}_v)] - \sum_{i=1}^{k} \log[\sigma(-\mathbf{z}_u^T \mathbf{z}_{n_i})]$$

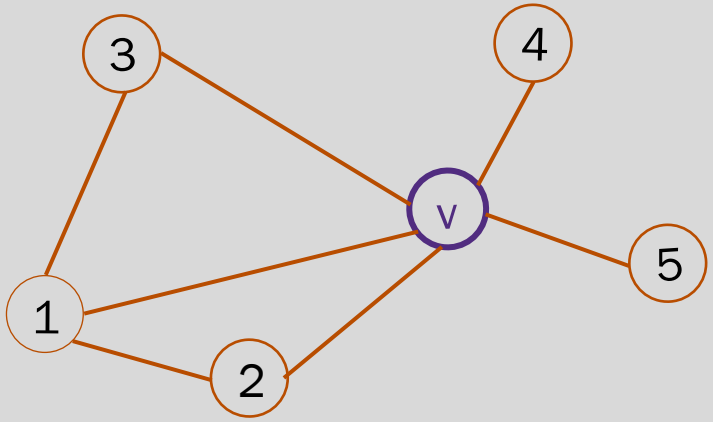
What is a random walk policy?

- As of yet, we haven't specified the random walk policy R
- Random walk policy definition
  - Assume the random walk is at node *v*
  - *Set a rule (policy) for deciding which neighbor node of v to select for the next step*
  - The rule can depend on parameters and other information about the walk state and network



How do we pick the random walk policy?

- Simplest approach is to select the next node in the walk from the neighbors of the current node with equal probability
  - Unbiased random walk
  - This was proposed for DeepWalk by Perozzi et al.
  - This turns out to be too restrictive
- How can we improve this?



Unbiased Random Walk Policy Example

Node *v* has 5 neighbors. Pick a number, ρ, from a uniform random distribution on [0,1]. Pick the next node, u, with the rule:

$\rho \in [0, 0.2) \longrightarrow u = 1$
$\rho \in (0.2, 0.4) \longrightarrow u = 2$
$\rho \in (0.4\ 0.6) \longrightarrow u = 3$
$\rho \in (0.6, 0.8) \longrightarrow u = 4$
$\rho \in (0.8, 1] \longrightarrow u = 5$

Breadth and depth first search

- **Breadth-first** is a walk policy that favors a local neighborhood view
- **Depth-first** is a walk policy that favors a global neighborhood views
- We would like a walk policy that combines these
- Next, we discuss *node2vec* which uses a parameterized bias random walk policy that interpolates between **breadth-first (local)** and **depth-first (global)** views of a node neighborhood

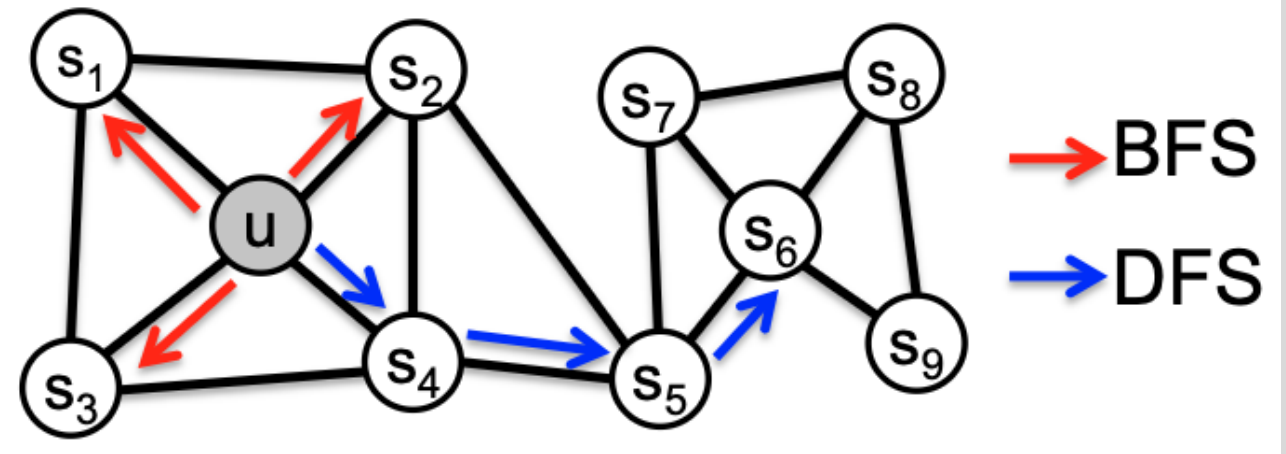
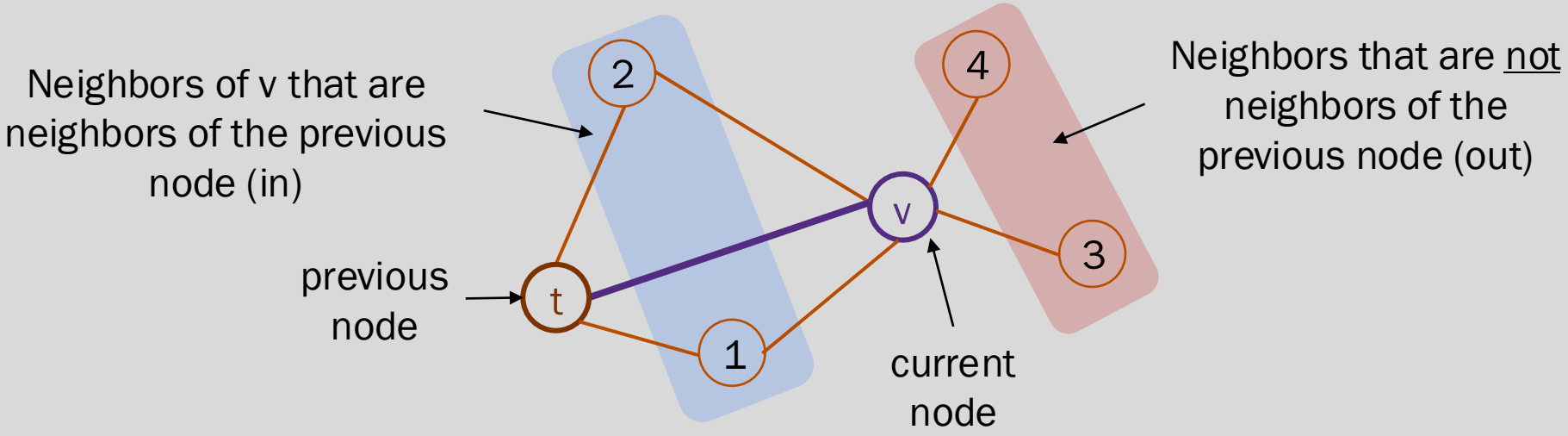


Image credit: Grover & Leskovec 2016

node2vec uses a biased random walk policy

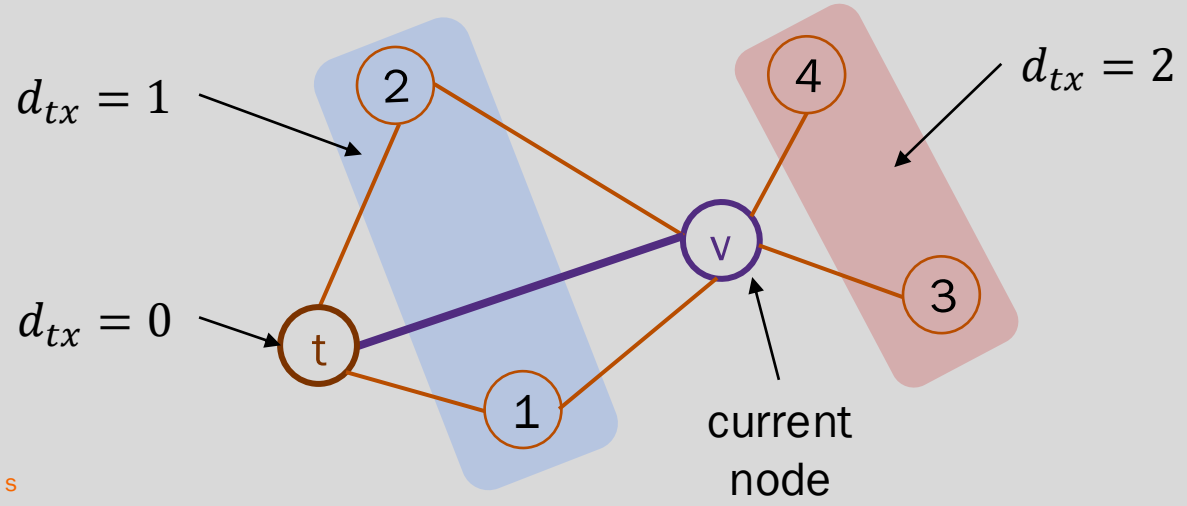
- Biased random walk concept
  - Assume the random walk just traversed the edge from node *t* to node *v*
  - **Bias the probability of picking a neighbor of node v based on whether it is**
    - The previous node
    - A neighbor of the previous node
    - Not a neighbor of the previous node
  - How do we bias the probability?



node2vec uses a biased random walk policy

- Biased probabilities
  - Assume the random walk just traversed the edge from node t to node v
  - Select policy parameters p and q (more on these later)
  - For node x in set of neighbors of v
    - Let w_{vx} be the weight on the edge between nodes v and x (unweighted $w_{vx} = 1$)
    - Let d_{tx} be the shortest path distance from node t to node x (d_{tx} is 0, 1, or 2)
    - Set the unnormalized transition probability to $\pi_{vx} = \alpha_{pq}(t,x) \cdot w_{vx}$
  - Normalize the transition probabilities by $\sum_{x \in N(v)} \pi_{vx}$

$$\alpha_{pq}(t,x) = \begin{cases} \frac{1}{p} & \text{if } d_{tx} = 0 \\ 1 & \text{if } d_{tx} = 1 \\ \frac{1}{q} & \text{if } d_{tx} = 2 \end{cases}$$



node2vec biased random walk step example

- Let $\Pi_v = \sum_{x \in N(v)} \pi_{vx}$ be the sum of the unnormalized transition probabilities $\pi_{vx} = \alpha_{pq}(t,x) \cdot w_{vx}$ so that the normalized transition probability from node v to x is $\psi_{vx} = {}^{\pi_{vx}}/_{\Pi_v}$.

Example

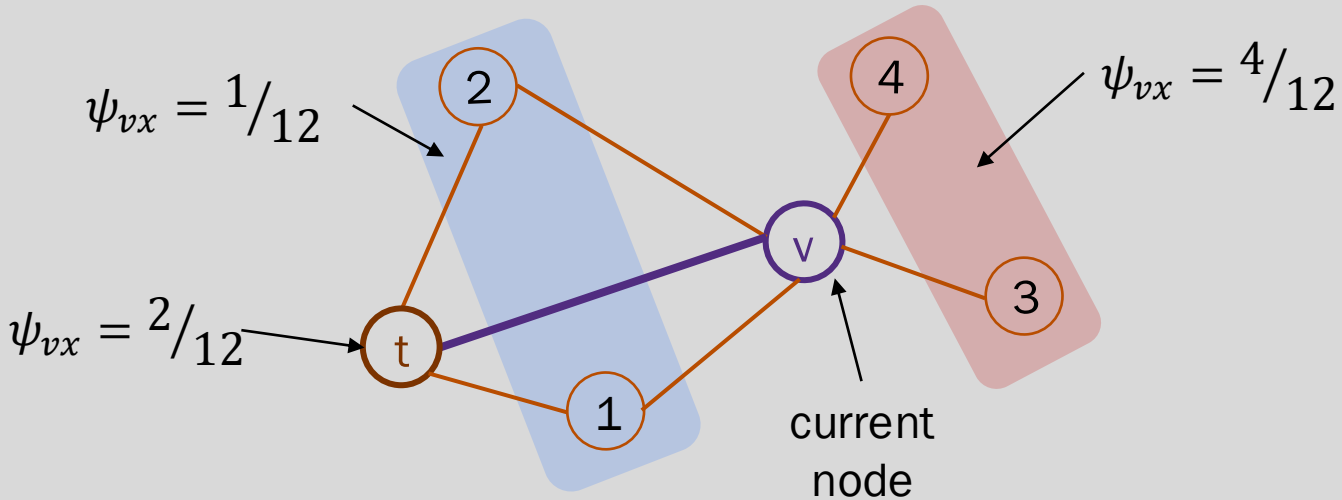
Let $w_{vx} = 1$ for all x (unweighted)

Let $p = 0.5$ and $q = 0.25$

$\pi_{vt} = 2, \pi_{v1} = \pi_{v2} = 1, \pi_{v3} = \pi_{v4} = 4$

$\Pi_v = \sum_{x \in N(v)} \pi_{vx} = 2 + 2 \cdot 1 + 2 \cdot 4 = 12$

$$\alpha_{pq}(t,x) = \begin{cases} \frac{1}{p} & \text{if } d_{tx} = 0 \\ 1 & \text{if } d_{tx} = 1 \\ \frac{1}{q} & \text{if } d_{tx} = 2 \end{cases}$$



Pick a number, ρ, from a uniform random distribution on [0,1]. Pick the next node, u, with the rule:

$\rho \in [0, {}^{1}/_{12}) \longrightarrow u = 1$

$\rho \in ({}^{1}/_{12}, {}^{2}/_{12}) \longrightarrow u = 2$

$\rho \in ({}^{2}/_{12}, {}^{4}/_{12}) \longrightarrow u = t$

$\rho \in ({}^{4}/_{12}, {}^{8}/_{12}) \longrightarrow u = 3$

$\rho \in ({}^{8}/_{12}, 1] \longrightarrow u = 4$

https://ml4g.masinolab.com/topics/shallow-embeddings/node2vec.html

node2vec uses a biased random walk policy

- Random walk policy parameters
  - In-out parameter q: scales the probability of staying in (BFS) the previous node's neighbors vs. moving out (DFS) of the previous node's neighbors (DFS)
  - Setting $q > 1$ encourages BFS and $q < 1$ encourages DFS (we saw this in the previous example)
  - Return parameter p: scales the probability of returning to the previous node in a walk
  - Setting $p > \max(q, 1)$ discourages backtracking, setting $p < min(q, 1)$ encourages backtracking, otherwise $min(q, 1) < p < \max(q, 1)$ has moderate backtracking

https://ml4g.masinolab.com/topics/shallow-embeddings/node2vec.html

node2vec context sampling

- Assume we have created a large number of random walks with the node2vec biased walk policy
- How do we use them to create samples?
- Consider a single random walk

1. Set the context window length w
2. Pick a center node, u, in the walk
3. Let every node, v, that is within w steps to u be a context node
4. Form positive pairs (u, v) for u and every context node
5. Form a single sample as the set of positive pairs and k randomly sampled negative nodes (i.e., those not in the graph that are not in the context nodes)

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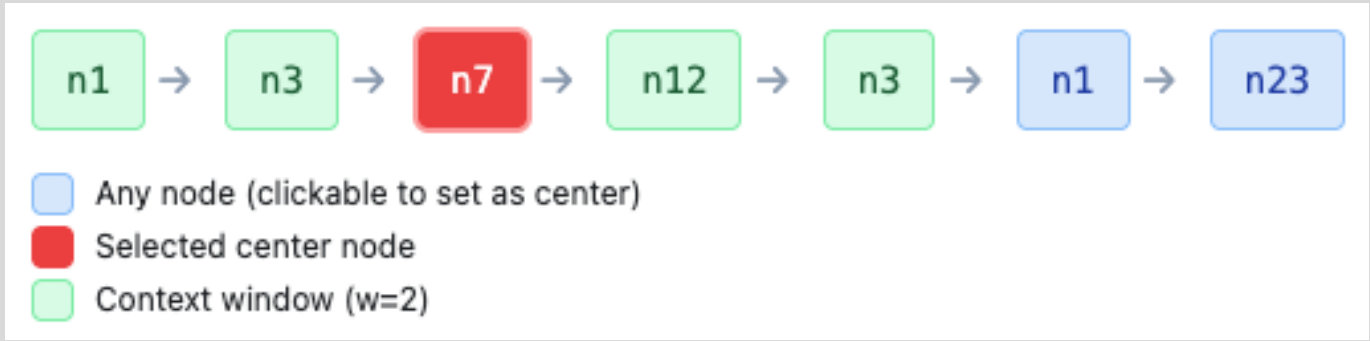
node2vec context sampling

A single random walk of length 6

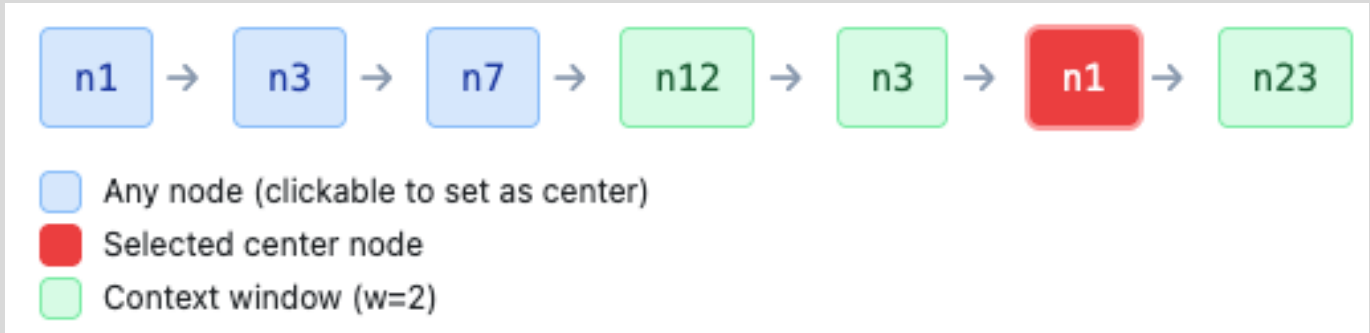


Context Window for center node n7 at position 2

Context nodes: `n1, n3, n12, n3`

Window range: `[0, 4]`

Positive pairs:
`(n7, n1), (n7, n3), (n7, n12), (n7, n3)`



Context Window for center node n1 at position 5

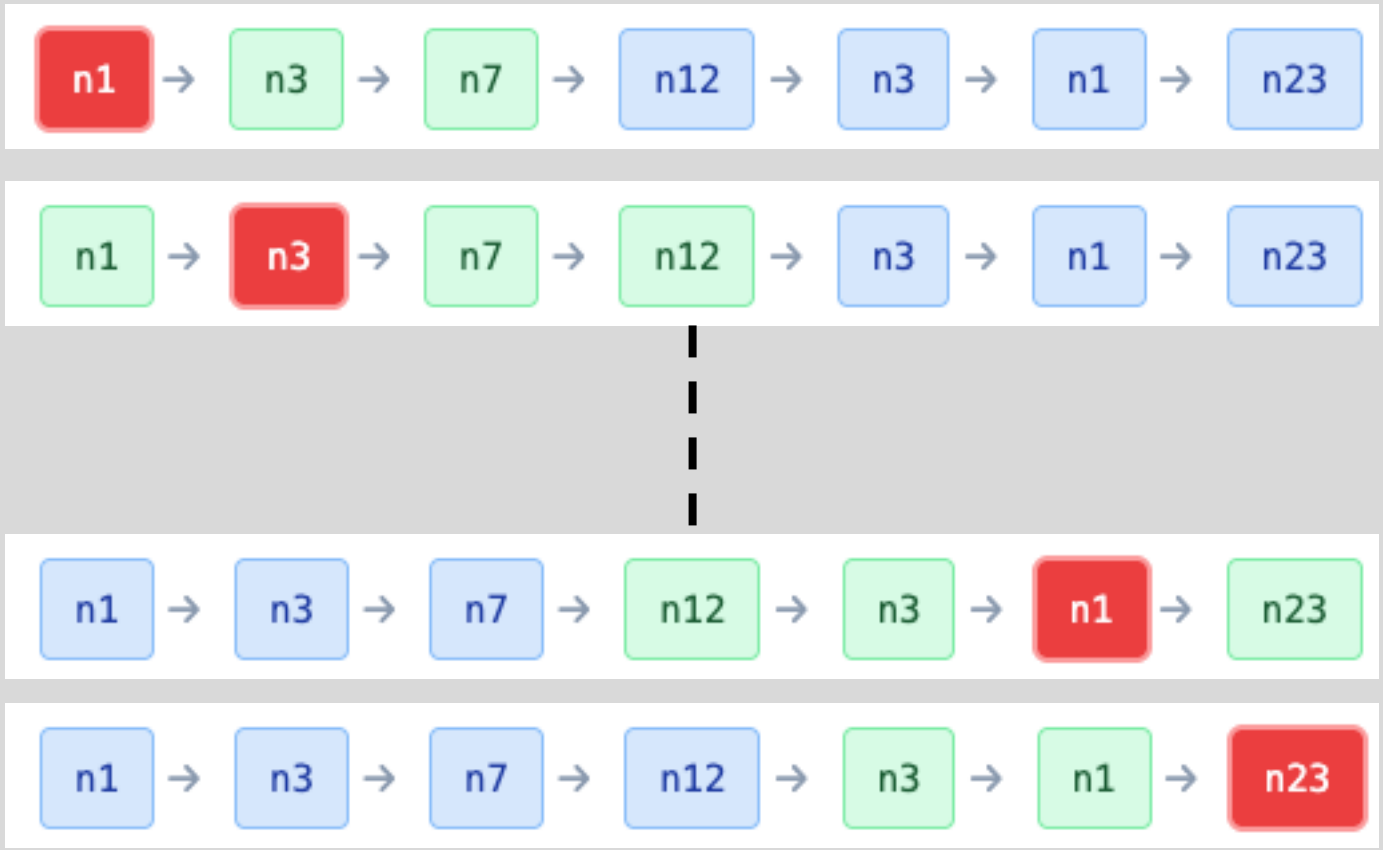
Context nodes: `n12, n3, n23`

Window range: `[3, 6]`

Positive pairs:
`(n1, n12), (n1, n3), (n1, n23)`

https://ml4g.masinolab.com/topics/shallow-embeddings/node2vec.html

node2vec context sampling



Each row is a single sample for SGD

Sample	Walk	Center Pos	Center Node	Context	Positive Pairs	Negative Samples
1	1	0	n1	n3, n7	(n1, n3), (n1, n7)	n45, n29, n86, n63, n24
2	1	1	n3	n1, n7, n12	(n3, n1), (n3, n7), (n3, n12)	n41, n86, n99, n72, n27
3	1	2	n7	n1, n3, n12, n3	(n7, n1), (n7, n3), (n7, n12), (n7, n3)	n14, n5, n81, n61, n66
4	1	3	n12	n3, n7, n3, n1	(n12, n3), (n12, n7), (n12, n3), (n12, n1)	n44, n68, n15, n81, n37
5	1	4	n3	n7, n12, n1, n23	(n3, n7), (n3, n12), (n3, n1), (n3, n23)	n74, n15, n28, n20, n19
6	1	5	n1	n12, n3, n23	(n1, n12), (n1, n3), (n1, n23)	n81, n2, n54, n87, n26
7	1	6	n23	n3, n1	(n23, n3), (n23, n1)	n83, n29, n27, n45, n87

https://ml4g.masinolab.com/topics/shallow-embeddings/node2vec.html

node2vec algorithm summary

1. Precompute edge transition probabilities for all edges
2. Generate r random walks of length l for every node in the graph using the bias random walk policy
3. Use context sampling to generate samples where each sample include positive pairs (u, v_w) and k negative nodes
4. Apply SGD over each sample using NCE with negative sampling loss

- Linear time complexity
- All steps are individually parallelizable

https://ml4g.masinolab.com/topics/shallow-embeddings/node2vec.html

Random walk variants

- Different kinds of biased random walks:
  - Based on node attributes (Dong et al.)
  - Based on learned weights (Abu-El-Haija et al.)
- Alternative optimization schemes:
  - Directly optimize based on 1-hop and 2-hop random walk probabilities (as in LINE from Tang et al.)
- Network preprocessing techniques:
  - Run random walks on modified versions of the original network (e.g., Ribeiro et al. struct2vec, Chen et al. HARP)

Summary so far

- Core idea: Embed nodes so that distances in embedding space reflect node similarities in the original network
- **Different notions of node similarity:**
  - Naïve: Similar if two nodes are connected
  - Random walk approaches

Summary so far

- So, what method should I use..?
- **No one method wins in all cases....**
  - E.g., node2vec performs better on node classification while alternative methods perform better on link prediction (Goyal and Ferrara, survey)
- Random walk approaches are generally more efficient
- In general: Must choose definition of node similarity that matches your application

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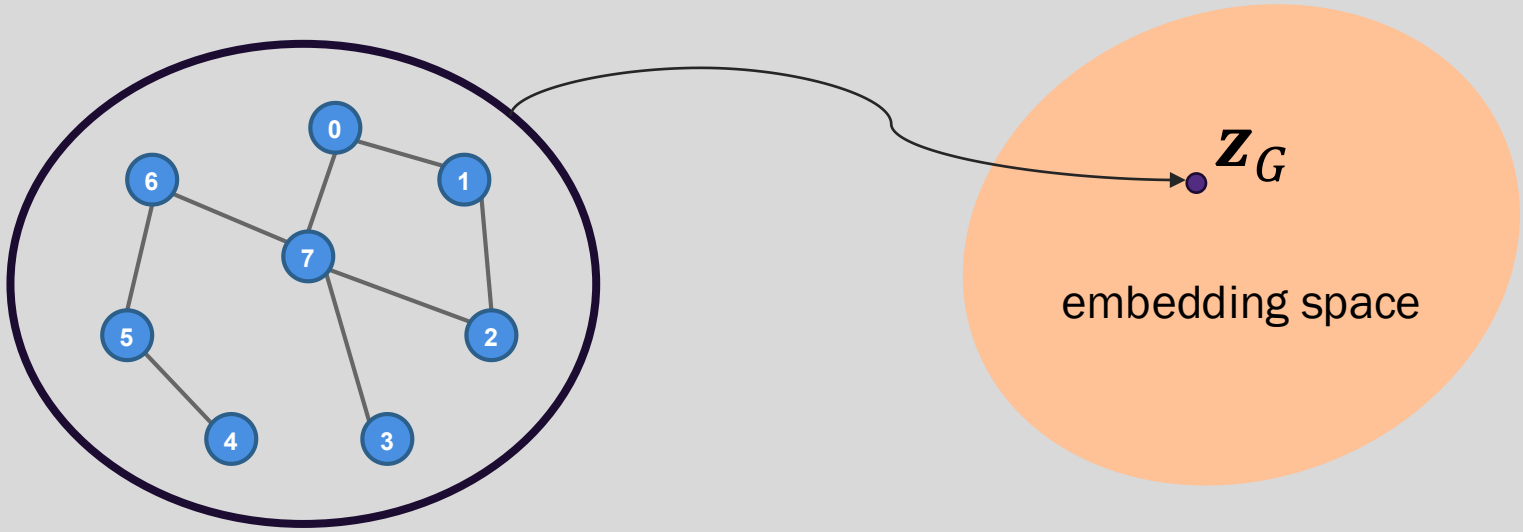
Graph Embeddings

Embedding an entire graph

Goal: Embed a subgraph (or the entire graph) G in the embedding space

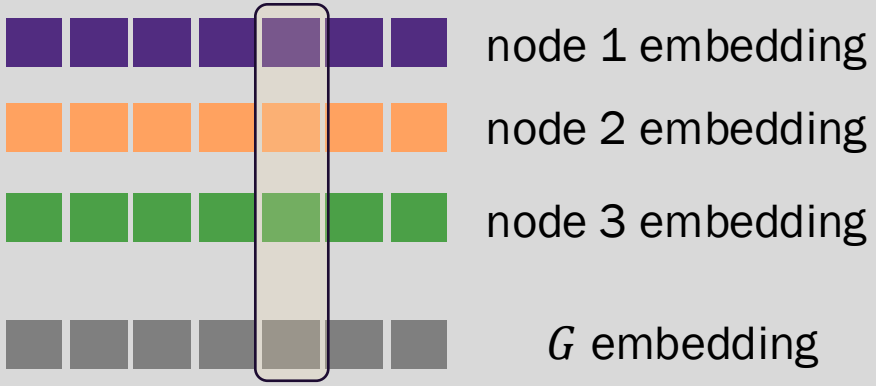
Motivation: Classify or cluster networks

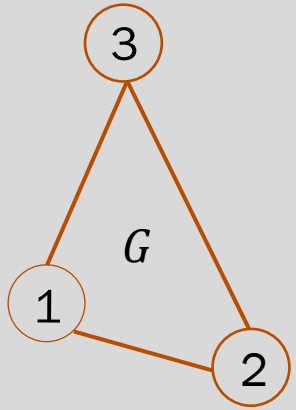
Examples: candidate drug targets, anomalous networks



Node averaging for graph embedding

- Apply a node embedding algorithm to all nodes in G
- Average the node embeddings to obtain a single embedding for G
- The average is done element wise

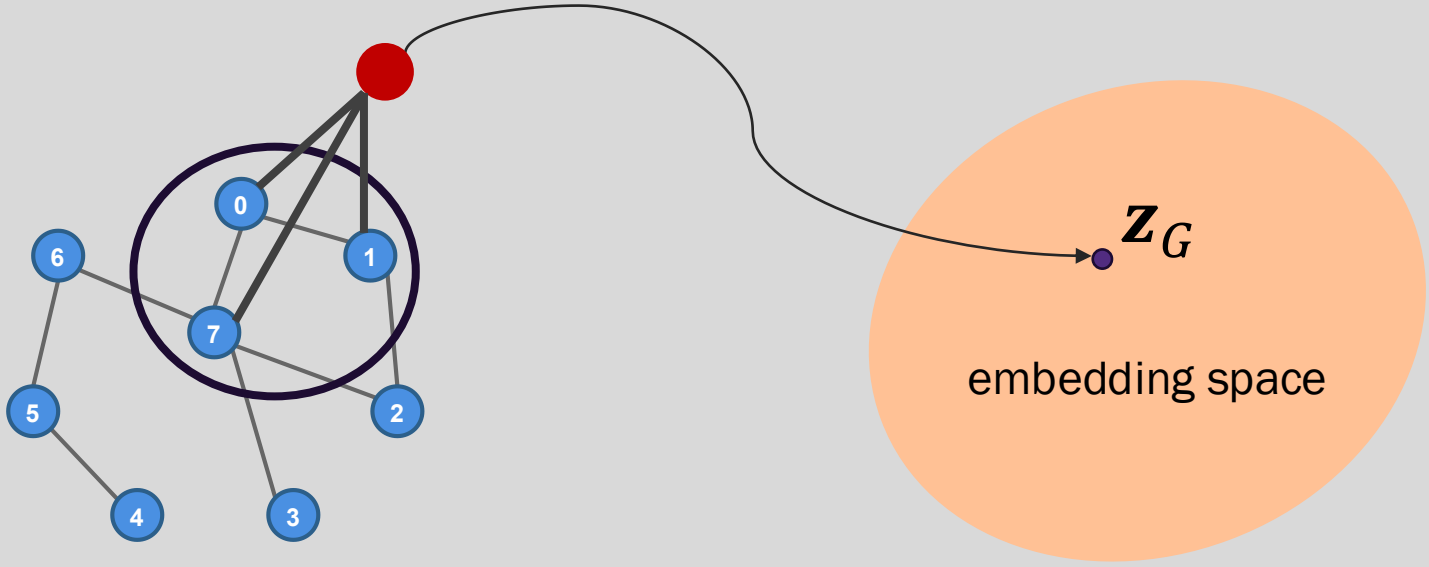




position *i* in G embedding is the average of the embedding values at position *i* from the node embeddings

Virtual nodes for graph embedding

- Create a virtual node and connect all nodes in the subgraph (or entire graph) to the virtual node
- Run a node embedding method to obtain the embedding for the virtual node



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Limitations of matrix factorization and random walk embeddings

Matrix factorization and random walk embedding methods are transductive

- Transductive (not inductive) method:
  - Cannot obtain embeddings for nodes not in the training set
  - Cannot apply to new graphs using the same nodes or the same graph with updated edges
    - If you apply a random walk method to the same graph multiple times, you'll get a different embedding each time. Why?

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Random walk embedding methods cannot capture structural similarity

- DeepWalk and node2vec do not capture structural similarity
- Consider the network below
  - Node 1 and 11 are structurally similar – part of one triangle, degree 2, …
  - However, they have very different random walk embeddings.
    - Why? It's unlikely that a random walk will reach node 11 from node 1

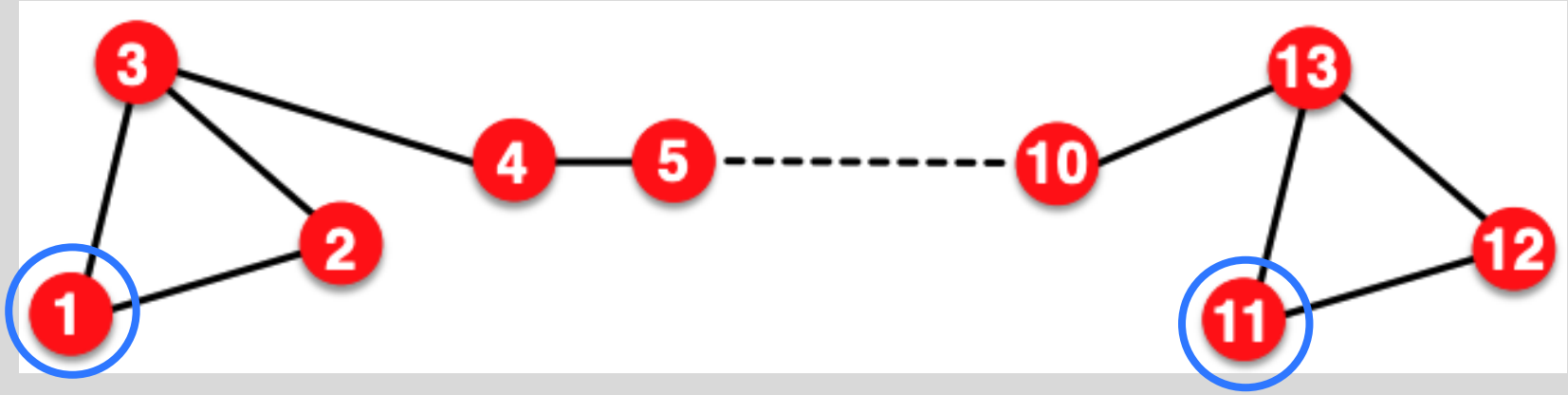

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Matrix factorization and random walk embedding methods cannot use node, edge, or graph features

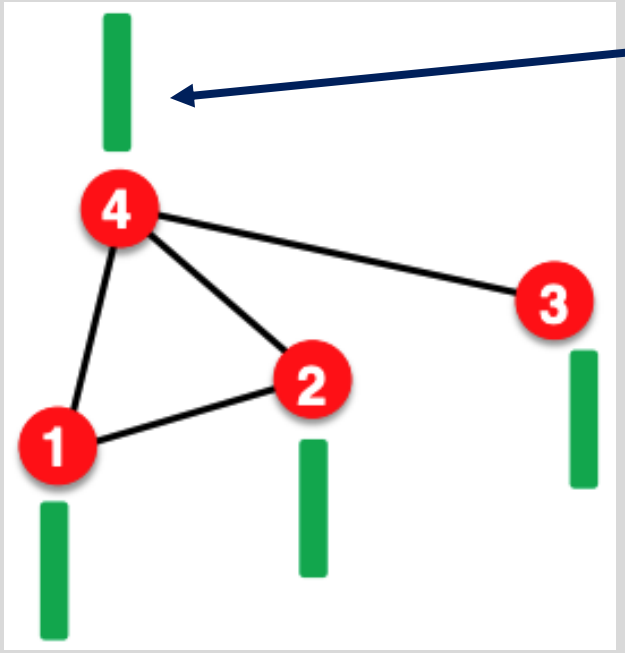

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Node feature vector
(e.g. protein properties in a protein-protein interaction graph)

Matrix factorization / random walk embeddings do not incorporate such node features

Solution: Graph neural networks

What can we do with node embeddings?

- Clustering/community detection: Cluster points $\mathbf{z}_n$
- Node classification: Predict label of node n based on $\mathbf{z}_n$
- Link prediction: Predict edge (u, v) based on a function of $f\big(g(\mathbf{z}_u, \mathbf{z}_v)\big)$ where the function g is usually on of vector concatenation, averaging, Hadamard product, or difference
- Graph classification: Predict label based on graph embedding, $\mathbf{z}_G$, formed via aggregating node embeddings or virtual-node